Layout-Controlled Synthetic Data Generation for Remote Sensing Object Detection

Yu Qiao, Xiang Zhang, *Member, IEEE*, Chenji Wang, Shengmei Chen, *Graduate Student Member, IEEE*, Yanzi Shi, *Member, IEEE*, Hangzai Luo, and Sheng Zhong

***Abstract*—High-performance remote sensing object detection typically requires vast datasets with annotations, but the collection and labeling of large-scale data is a costly and labor-intensive process. While generative models have shown promise in addressing the need for large-scale datasets, they heavily rely on high-quality labeled data and often overlook crucial aspects, such as the interobject relationships and overall quality of the synthetic data. To this end, we propose layout-controlled synthetic data generation (LSDGen), a novel approach for generating high-fidelity remote sensing imagery for object detection. LSDGen consists of three key stages: 1) relation-constrained layout generation (RLGen), which constructs a category association matrix from benchmark data to generate both main and subcategories, as well as their spatial configurations, without relying on bounding-box annotations; 2) image synthesis via a layout-conditional diffusion model (LCDM), where the generated layout guides a diffusion model to produce high-quality remote sensing images; and 3) instance-centered synthetic data filtering (ISDF), which evaluates the semantic consistency of individual instances in the synthesized images by calculating instance-level cosine similarity scores between visual and textual representations, filtering out low-confidence samples based on the average semantic score. Experimental results on the object DetectIon in Optical Remote sensing images (DIOR), oriented object detection benchmark based on DIOR dataset (DIOR-R), and high-resolution ship corpus (HRSC) datasets demonstrate that our method significantly improves the detection performance of existing remote sensing image object detection models, outperforming state-of-the-art annotation-dependent synthesis approaches.**

***Index Terms*—Object detection, remote sensing images, synthetic data filtering, synthetic data generation.**

I. Introduction

REMOTE sensing image object detection is crucial for applications such as urban planning, agricultural monitoring, and disaster assessment, enabling the automated localization and recognition of objects in complex aerial imagery [1], [2]. Despite significant advances in detection methodologies [3], [4], [5], their effectiveness is constrained by the dependence on large-scale, high-quality annotated datasets. Remote sensing imagery, with its complex semantic structures and densely clustered targets, poses significant challenges for accurate data annotation. Manual annotation is labor-intensive, time-consuming, and prone to biases and inconsistencies, limiting the scalability and effective deployment of object detection systems in real-world applications.

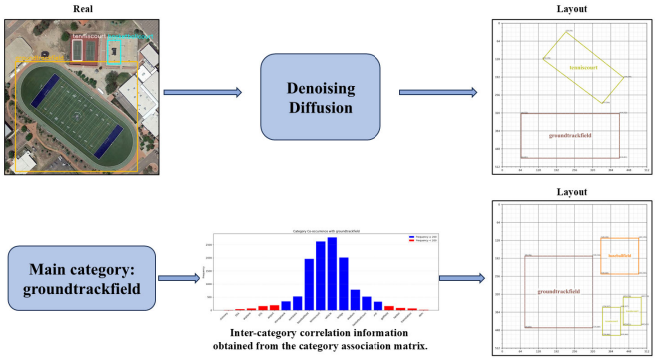


Fig. 1. Comparison of layout generation methods between AeroGen [2] and our LSDGen. The top row presents examples from AeroGen, which depends on manually annotated bounding boxes, while the bottom row showcases results from LSDGen, which generates layouts purely based on intercategory correlation, eliminating the need for manual annotation of bounding boxes.

To alleviate the heavy burden of data collection and annotation, various remote sensing data augmentation strategies have been widely employed in expanding training datasets for object detection, such as using rotation, scaling, and other traditional augmentation methods [6]. While these methods can increase the volume of data, the augmented data often lacks sufficient diversity, failing to meet the model's need for complex scenes. Additionally, when using the copy–paste strategy [7] for data augmentation, the pasted instances often suffer from semantic misalignment with the background regions. For example, ships may appear on roads or buildings instead of water, violating spatial and contextual coherence, which further reduces the effectiveness of synthetic data in model training.

Recent advancements in diffusion models have significantly impacted the field of controllable image synthesis, especially within the context of remote sensing applications. Two dominant paradigms have emerged: text-to-image (T2I) [8], [9] and layout-to-image (L2I) [10], [11]. T2I methods offer a versatile interface for describing scene content, enabling the generation of visually plausible remote sensing images from textual prompts. However, T2I models often struggle to generate images with precise bounding-box annotations. On the other hand, L2I methods condition image generation

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on predefined layouts, thereby providing explicit control over object categories, spatial positions, and sizes. This feature makes the L2I particularly effective when generating data with bounding-box annotations. However, current L2I methods rely on the spatial configurations of instances in the original dataset, resulting in generated layouts that closely resemble the original data (see Fig. 1), thereby restricting the diversity of generated data and hindering their application in complex remote sensing scenarios.

Based on the above observations, we introduce layout-controlled synthetic data generation (LSDGen), a novel framework designed to create large-scale, high-quality synthetic data. LSDGen consists of three key components: 1) relation-constrained layout generation (RLGen), which leverages category relationship information from benchmark datasets to construct a category association matrix, guiding the selection of main and semantically relevant subcategories, as well as their spatial placement in the layout, all without relying on bounding-box annotations from real data; 2) image synthesis via layout-conditional diffusion model (LCDM), where the generated layouts serve as structured input for an LCDM, producing high-resolution remote sensing images that respect the specified layout; and 3) instance-centered synthetic data filtering (ISDF), which employs the CLIP model [12] to compute cosine similarity between the adjudicated instances and their corresponding textual embeddings, deriving an overall average semantic score to filter out synthetic samples with low semantic scores.

The key contributions of this work are as follows.

1) We introduce a novel framework, RLGen, which exploits relational constraints between remote sensing object categories to generate semantically consistent and diverse layouts.
2) An ISDF method is developed to purify the quality of synthetic data. It improves synthetic data at the instance level by utilizing the CLIP model.
3) Experimental results show that our LSDGen substantially enhances the detection performance of existing remote sensing image object detection models, yielding improvements in mAP by 3.3%, 3.5%, and 6.7% on the object DetectIon in Optical Remote sensing images (DIOR), oriented object detection benchmark based on DIOR dataset (DIOR-R), and high-resolution ship corpus (HRSC) datasets, respectively.

II. Related Works

A. Object Detection in Remote Sensing Images

Recent advancements in object detection algorithms have significantly enhanced the analysis capabilities of remote sensing imagery. Two-stage approaches like Fast R-CNN [13] and Faster R-CNN [14] have shown strong performance in detecting small, densely distributed objects by decoupling region proposal and classification processes. On the other hand, one-stage detectors such as the You Only Look Once (YOLO) family [15], [16] are known for their fast inference speed and competitive accuracy, making them particularly suitable for large-scale and real-time remote sensing tasks, including disaster monitoring and rapid urban change detection. Although these methods have achieved good performance, their success is largely owed to the availability of large-scale training images with bounding-box annotations. In this work, we focus on the generation of large-scale data with bounding-box annotations to augment training data, thereby improving the performance of existing remote sensing image object detection models.

B. Data Generation and Augmentation

Early data augmentation relied on traditional techniques such as geometric transformations [17], [18] and copy–paste [19]. However, these methods generally exhibit limited generative diversity and may produce images with semantic inconsistencies or misaligned foreground and background elements. Beyond these classical strategies, broader families of conditional generation approaches have also been explored in the literature. Image-to-image methods [20] demonstrate the ability to convert one visual domain into another with high fidelity, yet they often depend on paired data or strong structural correspondences and therefore provide limited flexibility in controlling multiobject spatial distributions. Methods [21] that synthesize images from segmentation introduce dense semantic guidance, but their applicability to remote sensing is constrained by the need for large quantities of accurate pixel-level annotations. Approaches [22] conditioned on structural cues such as pose offer improved spatial coherence but are typically developed for video settings rather than large-scale aerial imagery with diverse object arrangements. Style-guided generative models [23] allow fine-grained control of appearance, although such control alone does not ensure correct instance semantics or spatial configurations required for object detection tasks.

Recently, diffusion models [24], [25] have demonstrated strong capabilities in generating diverse and higher fidelity data. It has facilitated the development of two advanced controllability frameworks: T2I [8], [9] and L2I [10], [11]. T2I grants semantic-level flexibility through linguistic prompts but demonstrates limited spatial precision in multiobject scenes. L2I rectifies spatial incoherence through explicit instance spatial positioning, yet its dependence on manually curated layouts undermines scalability. Recent diffusion models (e.g., AeroGen [2] and SatSynth [26]) mitigate annotation burdens by learning synthetic layouts from preexisting datasets. These approaches are limited by the spatial configurations of instances in the original dataset, restricting the diversity of generated layouts and hindering their applicability in dynamic remote sensing scenarios. In this article, we propose LSDGen, consisting of three components: RLGen for layout construction using category relationships, layout-conditional diffusion to synthesize high-resolution images based on these layouts, and ISDF to filter low-confidence images by evaluating semantic consistency through cosine similarity between visual and textual representations.

III. Methodology

A. Overview

The proposed LSDGen framework, as illustrated in Fig. 2, is designed to generate high-quality remote sensing images

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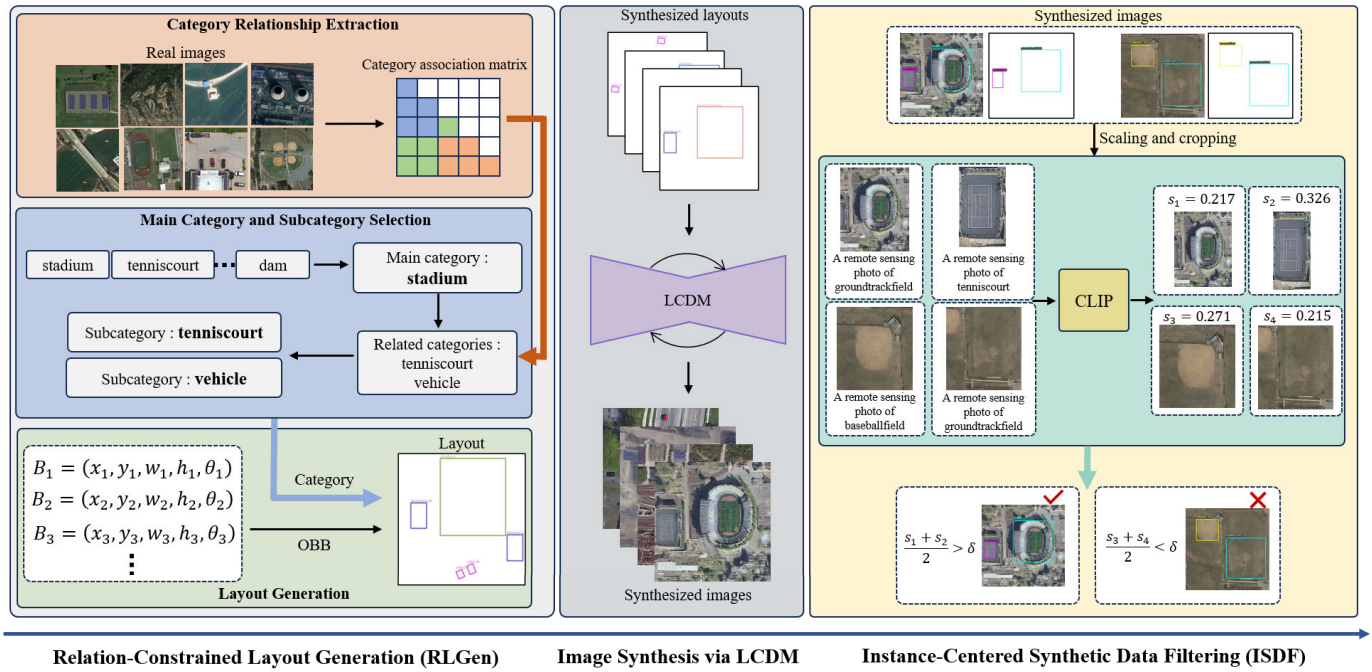


Fig. 2. Overview of the LSDGen. LSDGen comprises three main stages. It first employs RLGen, which constructs a category association matrix derived from real-world datasets to guide the generation of main and semantically related subcategories, together with their spatial placement in the layout. Each instance, represented by an OBB, is incorporated into the layout. The generated layouts are then used as conditional inputs to the LCDM [2], which synthesizes remote sensing imagery according to the specified layout. Finally, ISDF evaluates the semantic consistency of each synthesized instance by comparing its visual and textual embeddings, filtering out low-confidence synthetic samples.

with bounding-box annotations for object detection. LSDGen comprises three key modules: RLGen leverages category relationships from benchmark datasets to create layouts with randomly assigned categories and spatial configurations. The LCDM then synthesizes high-resolution images based on these layouts. Finally, ISDF assesses the semantic alignment of synthesized objects by measuring their visual–textual similarity, filtering out instances with low alignment to ensure high-quality data.

B. Relation-Constrained Layout Generation

Many existing layout-based image generation models, such as AeroGen [2], rely heavily on extensive manual bounding-box annotations, which constrain the creativity and diversity of the generated layouts. This limitation makes them less suitable for the dynamic and diverse nature of real-world remote sensing scenarios. In contrast, fully random layout generation eliminates this dependency but often introduces semantic and spatial inconsistencies. To this end, we propose RLGen, which constructs semantically consistent layouts by modeling intercategory relationships through a category association matrix, enabling layout generation without relying on real bounding-box annotations. RLGen first excavates the category relationships in the benchmark data to construct a category association matrix, which serves as the prior knowledge. Based on this matrix, the main category is randomly generated, and the corresponding subcategories are determined according to the matrix, serving as the filling categories in the layout. For the generation of spatial positions, a randomization strategy is employed to determine the precise locations of the objects in the layout.

1) Category Relationship Extraction: We first extract inter-category correlation information from real remote sensing data and represent it as a category association matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$, defined as

$$\mathbf{A} = \begin{bmatrix} \rho(c_1,c_1) & \rho(c_1,c_2) & \cdots & \rho(c_1,c_N) \\ \rho(c_2,c_1) & \rho(c_2,c_2) & \cdots & \rho(c_2,c_N) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(c_N,c_1) & \rho(c_N,c_2) & \cdots & \rho(c_N,c_N) \end{bmatrix}$$

$$\text{where} \quad \rho(c_i,c_j) = \frac{f(c_i,c_j)}{\sum_{k=1}^{N} f(c_k,c_j)}. \tag{1}$$

Let N denote the total number of object categories in the dataset. The co-occurrence probability between categories c_i and c_j is represented as $\rho(c_i,c_j)$, where $f(c_i,c_j)$ denotes the number of co-occurrences between categories c_i and c_j, and $\sum_{k=1}^{N} f(c_k,c_j)$ represents the total number of co-occurrences of category c_j with all categories. The matrix $\mathbf{A}$ serves as prior knowledge to guide and constrain the selection of subcategories during the layout generation process.

2) Main Category and Subcategory Selection: The main category c_m is selected at random from the complete set of object categories $\mathcal{C} = [c_1, c_2, \ldots, c_N]$. Specifically, $m \sim \text{randint}\{1, N\}$, where m is the index of the selected category, drawn randomly from integers between 1 and N. This can be implemented using Python's `randint` function, which generates a random integer from a specified inclusive range.

To accurately model the object distribution in real-world scenarios, we determine the subcategories in the scene based

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on the selected main category c_m and the category association matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$. Specifically, we assume that, in addition to the main category, there are $M \in \{1, 2, \ldots, 8\}$ subcategories in the scene, where M is randomly selected from this range. The subcategories $C_s = \{c_s^1, c_s^2, \ldots, c_s^q, \ldots, c_s^M\}$ are selected at random from the candidate list $\mathcal{C}_{\text{sub}}$, which is constructed as

$$\begin{aligned} q &\sim \text{random.choice}\left(\mathcal{C}_{\text{sub}}\right) \\ \text{where} \quad \mathcal{C}_{\text{sub}} &= \text{Sort}\left(\{c_i \in \mathcal{C} \,|\, \rho\left(c_i, c_m\right) > \tau\}\right) \end{aligned} \tag{2}$$

where `random.choice` is used to randomly select an index q from the list $\mathcal{C}_{\text{sub}}$, thereby obtaining the subcategory c_s^q. Since M subcategories are generated in total, this process is repeated M times, resulting in the complete set of subcategories $C_s = \{c_s^1, c_s^2, \ldots, c_s^M\}$. $\rho(c_i, c_m)$ denotes the co-occurrence probability between category c_i and the main category c_m. The operation Sort$(\cdot)$ sorts the elements of the category association matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ that exceed the threshold τ (i.e., 0.05).

3) Layout Generation: After determining the object categories for a single layout, we generate an oriented bounding box (OBB) for each instance $i \in \{1, \ldots, M\}$. The OBB of the ith instance (i.e., B_i) is defined as follows:

$$\begin{aligned} B_i &= \left(x_i, y_i, w_i, h_i, \theta_i\right), \quad \theta_i \sim \text{randint}\left(0°, 90°\right) \\ \left(w_i, h_i\right) &= \alpha_i \cdot \left(\bar{w}_{c_i}, \bar{h}_{c_i}\right), \quad \alpha_i \sim \text{random.uniform}\left(0.5, 1.5\right) \\ \left(x_i, y_i\right) &\sim \text{random.uniform}\left([0, H] \times [0, W]\right) \\ \forall i \neq j, &\quad \text{IoU}\left(B_i, B_j\right) = 0 \end{aligned} \tag{3}$$

where (x_i, y_i) represents the center coordinates, while (w_i, h_i) are the width and height of the bounding box. $(\bar{w}_{c_i}, \bar{h}_{c_i})$ is the width and height of the object estimated based on human priors. The detailed parameters are provided in Section IV-A. The orientation angles θ_i are randomly chosen from 0° to 90° to simulate object rotations commonly seen in aerial imagery. After determining the width and height (w_i, h_i) as well as the orientation angle θ_i of each instance's bounding box, the spatial coordinates (x_i, y_i) for each instance are randomly assigned within the layout canvas. `random.uniform` is a Python function that returns a random floating-point number sampled from the given range. H and W represent the height and width of the layout. IoU(B_i, B_j) represents the intersection over union (IoU) of the ith instance and the jth instance. After obtaining all valid bounding boxes, we place the OBB of each instance into the layout. The resulting complete layout then guides the LCDM to synthesize remote sensing images.

C. Image Synthesis via LCDM

Recent studies have demonstrated that training L2I diffusion models can produce large-scale synthetic images. Motivated by this, we adopt the state-of-the-art LCDM [2] to generate remote sensing images for enhancing the performance of remote sensing image object detection models. Specifically, the layouts generated by RLGen, which consist of a set of OBBs, are used as conditional inputs for the LCDM to synthesize corresponding remote sensing images.

D. Instance-Centered Synthetic Data Filtering

Although diffusion models can generate visually realistic remote sensing imagery, they occasionally produce samples with semantic inconsistencies or structural distortions [27], [28] (see Fig. 3), which can lead to erroneous learning in downstream tasks due to the presence of misleading or inaccurate features. To this end, we propose an ISDF strategy, which leverages the CLIP model to compute the average semantic score for each synthesized image and filters out low-quality samples accordingly.

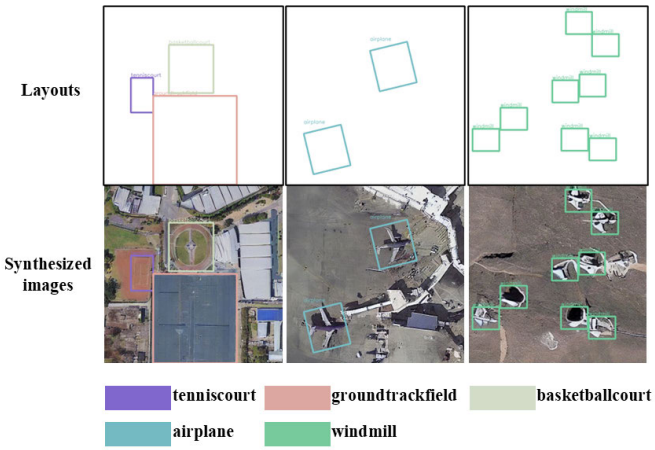


Fig. 3. Examples of synthetic data with low-fidelity instances.

While CLIP demonstrates strong zero-shot capabilities in assessing image–text alignment, it typically evaluates at the global image level, which contrasts with our focus on object instances. To address this issue, each instance in the synthetic image is cropped according to its corresponding bounding box, and the cosine similarity between all instances and their associated textual prompts within a single image is computed. If the average semantic score of the image falls below a predefined threshold, the entire synthetic image is discarded. Specifically, we first crop individual instances from the synthetic image. Given the varying semantic relevance between object categories and their surrounding backgrounds, we adopt a unified scaling strategy to ensure that each instance retains sufficient contextual background information. Each object is cropped from the synthesized image based on its layout annotation, with an extended region that includes the surrounding context. Suppose an instance is represented by an OBB $B = (x, y, w, h, \theta)$. We perform axis-aligned cropping by enlarging the original bounding box centered at (x, y) to twice its width and height. Once the cropped instances are obtained, each instance is evaluated by the CLIP to assess its semantic consistency with the corresponding category text prompt. To be specific, given a text prompt T_i in the form of *a remote sensing photograph of [category]* and an image instance X_i, their feature embeddings are extracted using the CLIP text encoder f_θ and the image encoder g_θ, respectively, resulting in $\mathbf{F}_{T_i} = f_\theta(T_i)$ and $\mathbf{F}_{X_i} = g_\theta(X_i)$. The average semantic score S of the image is computed as

$$S = \frac{1}{Z}\sum_{i=1}^{Z} s_i, \quad \text{where} \quad s_i = \frac{\mathbf{F}_{T_i}\mathbf{F}_{X_i}^{\top}}{\left\|\mathbf{F}_{T_i}\right\| \cdot \left\|\mathbf{F}_{X_i}\right\|} \tag{4}$$

where s_i denotes the cosine similarity score between the ith instance and its corresponding text prompt, where $\top$

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TABLE I
WIDTH AND HEIGHT OF OBJECTS IN EACH CATEGORY

Category	$\bar{w}_{c_i}$	h_{c_i}	Category	$\bar{w}_{c_i}$	h_{c_i}
airplane	120	110	airport	400	200
baseballfield	130	128	basketballcourt	128	64
bridge	190	120	chimney	210	200
dam	350	260	Expressway-Service-area	350	310
Expressway-toll-station	100	90	golffield	410	390
groundtrackfield	240	256	harbor	300	200
overpass	310	200	ship	256	64
stadium	260	300	storagetank	68	64
tenniscourt	64	100	trainstation	360	200
vehicle	36	26	windmill	76	64

denotes the transpose operator and $\|\cdot\|$ represents the ℓ_2-norm. The final average semantic score S is the average over all Z instances in the synthesized image. The image is retained if $S \geq \delta$, otherwise discarded. In this article, we set $\delta = 0.25$ in all experiments.

IV. EXPERIMENTAL

A. Experimental Setup

1) DIOR, DIOR-R, and HRSC: In this paper, we conduct experiments on three widely adopted benchmark datasets: DIOR [29], DIOR-R [30], and HRSC [31]. DIOR and DIOR-R share the same set of 23 463 images containing a total of 1 92 472 object instances across 20 categories. They differ in annotation format: DIOR employs horizontal bounding boxes (HBBs), while DIOR-R utilizes OBBs. Both datasets are split into training, validation, and test sets in a 1 : 1 : 2 ratio. The HRSC dataset is divided into 436 training images, 181 for validation, and 444 for testing.

2) Syn-DIOR, Syn-DIOR-R, and Syn-HRSC: To enhance the performance of remote sensing object detection, we construct three synthetic datasets: Syn-DIOR, Syn-DIOR-R, and Syn-HRSC. Syn-DIOR and Syn-DIOR-R share the same set of 50 000 images covering 20 object categories, but differ in their annotation formats. Syn-DIOR uses HBB, while Syn-DIOR-R adopts OBB. Syn-HRSC contains 10 000 images with a single object category. Table I shows the specific values of $(\bar{w}_{c_i}, \bar{h}_{c_i})$ for each category c_i. $(\bar{w}_{c_i}, \bar{h}_{c_i})$ represents the object's width and height derived from human prior knowledge, which is obtained via empirical observation of data. Fig. 4 shows several examples of synthesized images. It can be seen that the generated scenes preserve the semantic consistency characteristic of real remote sensing imagery, while also exhibiting greater diversity in the spatial arrangements of object instances.

3) Evaluation Metrics: For the quantitative evaluation of generated images, we employ the FID score [32] to assess visual quality. Furthermore, inspired by prior work [2], we adopt standard object detection metrics, including mAP@0.5 (mAP50) and mAP@0.5:0.95 (mAP), to evaluate the performance of object detection on the DIOR, DIOR-R, and HRSC *test* sets.

4) Implementation Details: In our experiment, the models are trained on an NVIDIA RTX 3090 GPU using PyTorch. During the training of the LCDM, we follow the configurations of AeroGen [2] and train the model separately on each dataset for 100 epochs using the AdamW optimizer with a learning rate of $1e^{-5}$. For the object detection, we adopt YOLOv8 [15] as the object detector, where yolov8l is used for HBB detection and yolov8l-obb is employed for OBB detection. All models are trained for 50 epochs with a batch size of 16 using the SGD optimizer, a learning rate of 0.01, a momentum of 0.937, and a weight decay of 0.0005.

TABLE II
DETECTION PERFORMANCE OF YOLOV8 ON THE DIOR, DIOR-R, AND HRSC DATASETS UNDER VARYING SCALES OF SYNTHETIC DATA AUGMENTATION

Dataset	Synthetic Images	mAP	mAP50
DIOR	0	54.1	74.4
	10K	55.0	75.1
	20K	56.2	76.7
	50K	57.0	77.8
DIOR-R	0	49.3	66.3
	10K	51.3	67.3
	20K	52.0	68.1
	50K	52.5	68.5
HRSC	0	68.1	90.1
	2K	70.5	90.9
	4K	72.7	92.2
	10K	73.9	92.6

B. Ablation Studies

To evaluate the effectiveness of LSDGen, we conduct ablation experiments on the DIOR, DIOR-R, and HRSC datasets.

1) Effectiveness of Synthetic Images: We investigate the impact of augmentation quantity on performance. We use $10K$, $20K$, and $50K$ synthetic images to augment the training data of DIOR and DIOR-R datasets and use $2K$, $4K$, and $10K$ synthetic images to augment the training data of HRSC dataset. As shown in Table II, the performance of the models gradually improves as the number of synthetic images increases. This is attributed to our method, which can generate large-scale, high-quality, and diverse data, which effectively expands the training dataset and, in turn, improves the detector's generalization capability. Fig. 5 visually compares the per-class mAP of YOLOv8 on the DIOR-R dataset, with and without augmentation using 50K synthetic images.

2) Effectiveness of Synthetic Data Filtering: To evaluate the effectiveness of our ISDF, we progressively increase the amount of synthetic data used for augmentation. Specifically, synthetic datasets containing $10K$, $20K$, and $50K$ images are processed by ISDF, yielding filtered subsets of $8.3K$, $16.6K$, and $40.6K$ samples, respectively. Augmentation experiments are then conducted using these filtered datasets and compared with the corresponding unfiltered synthetic data. As shown in Table III, the model trained on data filtered using our ISDF improved detector performance, thereby validating the effectiveness of the proposed filtering approach in selecting high-quality synthetic data.

3) Effect of Threshold τ on Model Performance: To evaluate the influence of the threshold τ on the detection model, we conduct an ablation study. Since larger values of τ lead to a reduction in the number of synthesized images for rare categories, the threshold should remain below 0.10. Accordingly, we examine three representative values: 0.00, 0.05, and

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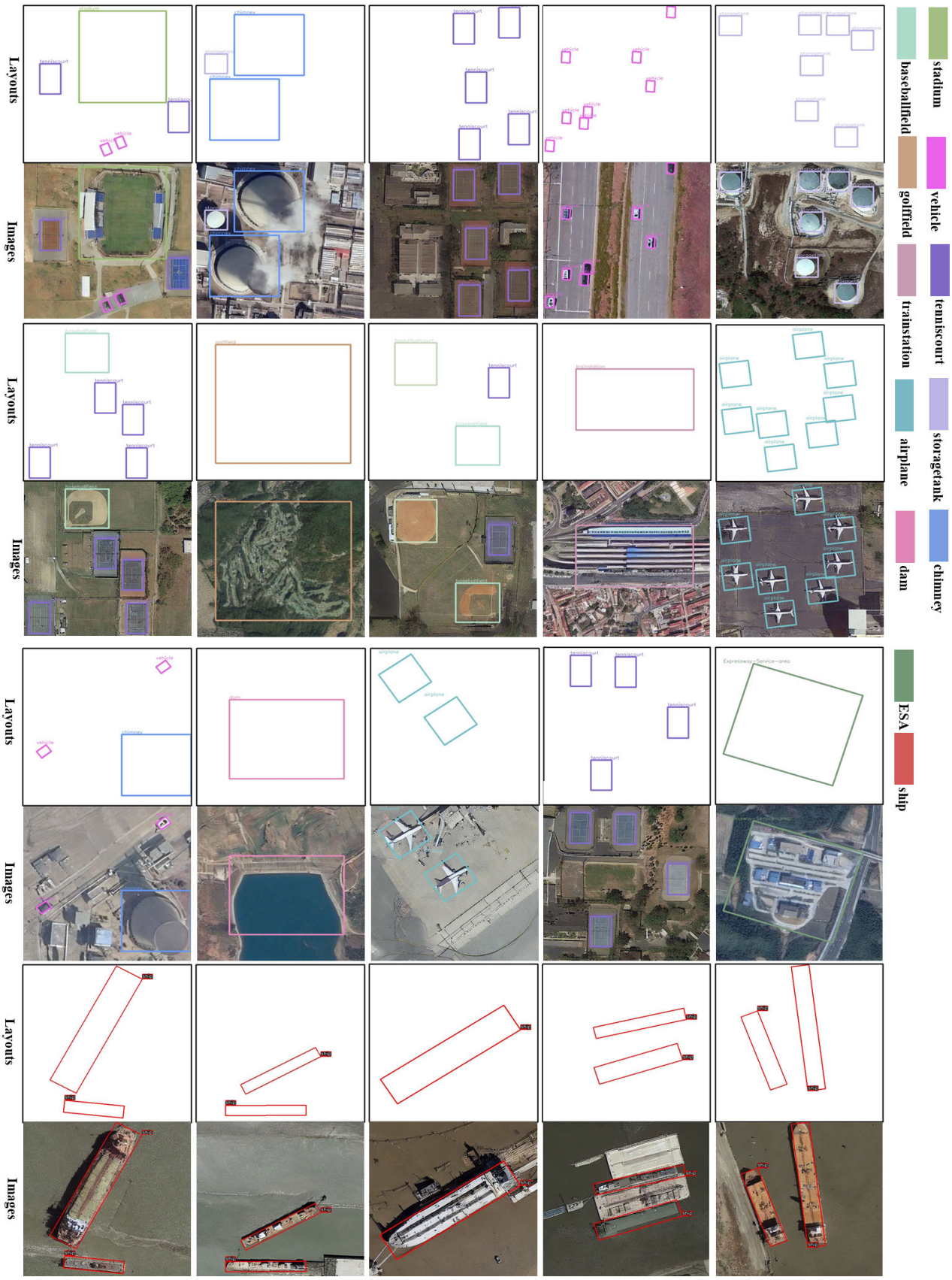


Fig. 4. Visualization of layouts generated by RLGen and the corresponding synthesized remote sensing images. The first three rows show samples from the Syn-DIOR and Syn-DIOR-R datasets, while the last row displays examples from the Syn-HRSC dataset.

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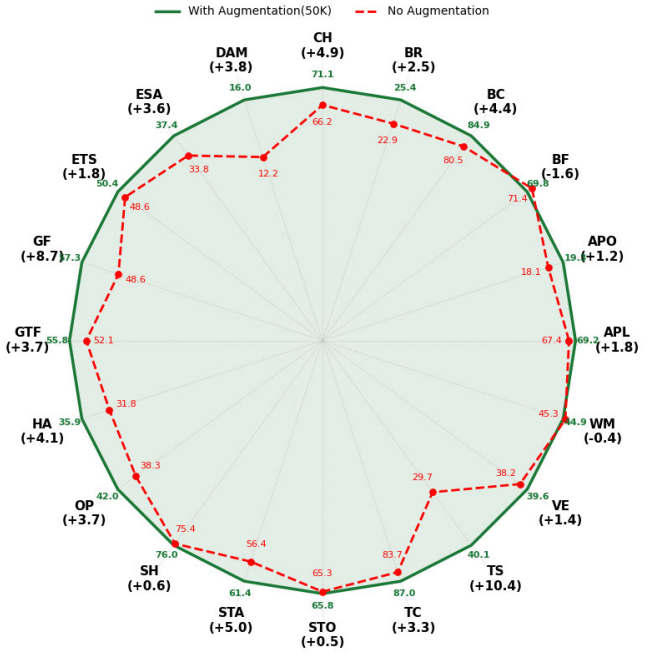


Fig. 5. Comparison of per-class mAP on the DIOR-R dataset with and without data augmentation.

TABLE III

ABLATION STUDY ON SYNTHETIC DATA FILTERING: DETECTION PERFORMANCE OF YOLOV8 WITH AND WITHOUT ISDF ON DIOR, DIOR-R, AND HRSC DATASETS

Dataset	Synthetic Images	mAP	mAP50
DIOR	10K (No Filter)	55.0	75.1
	8.3K (Filter)	55.3	75.5
	20K (No Filter)	56.2	76.7
	16.6K (Filter)	56.4	77.0
	50K (No Filter)	57.0	77.8
	40.6K (Filter)	57.4	77.7
DIOR-R	10K (No Filter)	51.3	67.3
	8.3K (Filter)	51.7	67.8
	20K (No Filter)	52.0	68.1
	16.6K (Filter)	52.4	68.5
	50K (No Filter)	52.5	68.5
	40.6K (Filter)	52.8	69.0
HRSC	2K (No Filter)	70.5	90.9
	1.8K (Filter)	71.4	91.3
	4K (No Filter)	72.7	92.2
	3.6K (Filter)	72.8	92.4
	10K (No Filter)	73.9	92.6
	9.1K (Filter)	74.8	93.0

TABLE IV

DETECTION PERFORMANCE ON THE DIOR-R DATASET USING DIFFERENT DETECTORS WITH AND WITHOUT 50K SYNTHETIC IMAGES

Detector	Synthetic Images	mAP50
YOLOv8	0	66.3
	50K	68.5
Oriented R-CNN	0	51.3
	50K	53.9

TABLE V

QUANTITATIVE RESULTS OF THE GENERATED IMAGES BY DIFFERENT METHODS

Method	Dataset	Modality	FID ↓
LostGAN [33]	DIOR	HBB	57.10
Layout Diffusion [10]			45.31
ReCo [34]			42.56
GLIGEN [35]			41.31
AeroGen [2]			38.57
LSDGen			**32.90**
GLIGEN [35]	DIOR-R	OBB	48.43
AeroGen [2]			35.07
LSDGen			**32.90**
GLIGEN [35]	HRSC	OBB	66.69
AeroGen [2]			45.86
LSDGen			**37.61**

TABLE VI

COMPARISON OF AUGMENTATION STRATEGIES ON THE DIOR-R DATASET WITH 50K SYNTHETIC IMAGES

Strategy	mAP	mAP50
Flip	37.4	60.2
CopyPaste	38.3	61.8
AeroGen	41.3	64.0
CopyPaste+Flip	38.8	62.1
AeroGen+Flip	41.7	64.1
LSDGen	**52.8**	**69.0**

TABLE VII

COMPARISON OF AUGMENTATION METHODS ON THE DIOR, DIOR-R, AND HRSC DATASETS

Dataset	Method	Gen Data	mAP	mAP50
DIOR	AeroGen	10K	55.6	74.8
		20K	56.8	76.3
		50K	57.9	77.1
	LSDGen	10K	55.3	75.5
		20K	56.4	77.0
		50K	57.4	77.7
DIOR-R	AeroGen	10K	39.8	62.4
		20K	41.1	63.3
		50K	41.7	64.1
	LSDGen	10K	51.7	67.8
		20K	52.4	68.5
		50K	52.8	69.0
HRSC	AeroGen	2K	64.1	91.8
		4K	64.8	92.3
		10K	65.9	93.1
	LSDGen	2K	71.4	91.3
		4K	72.8	92.4
		10K	74.8	93.0

TABLE VIII

IMPACT OF THE THRESHOLD τ ON DETECTION PERFORMANCE USING 10K SYNTHETIC IMAGES

Threshold τ	Dataset	mAP50	mAP
0.00	DIOR-R	67.0	51.5
0.05	DIOR-R	67.3	51.3
0.10	DIOR-R	66.8	51.2
0.00	DIOR	74.8	55.3
0.05	DIOR	75.1	55.0
0.10	DIOR	75.2	54.7

0.10. For each value of τ, we generate 10K synthetic samples and use them to augment the DIOR-R and DIOR datasets. As shown in Table VIII, the resulting detection performance remains almost unchanged across all settings, with only minor fluctuations in both mAP50 and mAP.

4) Effect of Threshold δ on Model Performance: To establish an appropriate operating range of δ, we first perform

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TABLE IX

COARSE-RANGE ANALYSIS OF THE ISDF THRESHOLD δ ON THE DIOR-R DATASET

Threshold δ	Synthetic Images	mAP50	mAP
0.150 (10k)	10K	67.3	51.3
0.250 (8.3k)	10K	**67.8**	**51.7**
0.350 (0.4k)	10K	66.5	49.7
0.450 (0k)	10K	66.3	49.3

TABLE X

FINE-RANGE ANALYSIS OF THE ISDF THRESHOLD δ ON THE DIOR-R DATASET ACROSS DIFFERENT SYNTHETIC DATA SCALES

Threshold δ	Synthetic Images	mAP50	mAP
0.245 (8.8k)	10K	66.9	51.3
0.250 (8.3k)	10K	**67.8**	**51.7**
0.255 (7.5k)	10K	66.4	51.1
0.245 (17.5k)	20K	67.9	52.4
0.250 (16.6k)	20K	**68.5**	**52.4**
0.255 (15.5k)	20K	67.6	52.1
0.245 (44.0k)	50K	67.9	52.5
0.250 (40.6k)	50K	**69.0**	**52.8**
0.255 (36.8k)	50K	68.0	52.6

a coarse-range evaluation using four representative values (0.150, 0.250, 0.350, and 0.450). As shown in Table IX, thresholds that are too low retain a substantial number of low-quality synthetic images, whereas excessively high thresholds discard nearly the entire dataset. A threshold around 0.250 yields a balanced compromise between filtering rigor and dataset preservation. Motivated by this observation, we perform a more refined study in Table X, where thresholds near 0.250 consistently achieve the best mAP50 and mAP scores across various synthetic data scales.

5) Analyzing the Impact of Detectors: To assess the generalization of the proposed method across different detectors, the performance of YOLOv8 [15] and Oriented R-CNN [36] is evaluated on the DIOR-R test set. Experiments are conducted using both the original DIOR-R dataset and the dataset augmented with 50K synthetic images. As reported in Table IV, the inclusion of synthetic data enhances the performance of both detectors. In addition, detectors with higher baseline capacity consistently achieve superior results, regardless of data augmentation.

6) Impact of Category Distribution on Model Performance: To examine how category distribution affects detection performance, we conducted additional experiments on both the DIOR-R and DIOR datasets. The results are summarized in Tables XI–XIV. We evaluated three training settings: 1) a baseline without synthetic augmentation; 2) training with synthetic augmentation; and 3) training with distribution-rebalanced synthetic augmentation. The rebalanced synthetic data increases the number of instances for rare categories, making it easier to assess whether additional samples for these categories result in performance improvements. However, some categories, such as *vehicle*, which frequently co-occur with other classes, remain relatively abundant even after rebalancing. Using the 10K synthetic data setting as an example, several consistent trends emerge. First, rare categories, such as *bridge* and *dam*, whose proportions in the real dataset

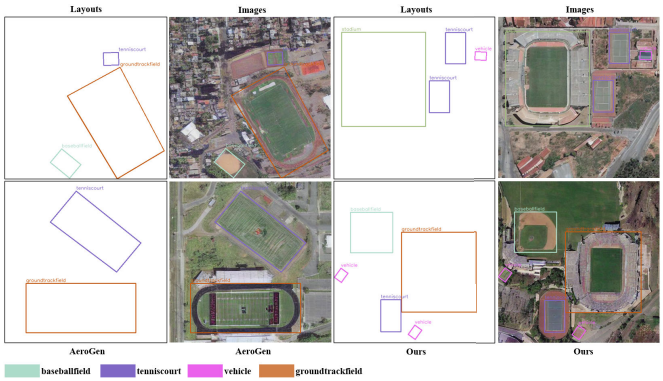


Fig. 6. Qualitative comparison of generated layouts and synthetic remote sensing images on the DIOR-R dataset.

fall below two percent, show clear performance improvements after augmentation. Second, further increasing the number of synthetic samples for these underrepresented categories results in additional gains. For example, the performance of *bridge* on DIOR-R improves from 39.7/24.1 to 42.3/26.4 in terms of mAP50/mAP. Third, challenging categories such as *harbor*, *golffield*, *expressway-service-area*, and *expressway-toll-station* also exhibit stable improvements in both mAP50 and mAP after synthetic augmentation and distribution rebalancing. These results demonstrate that the imbalance in the real dataset leads to reduced performance for rare categories in the detection model. In contrast, a balanced synthetic dataset significantly enhances model performance. Although our approach cannot guarantee an exactly consistent number of samples for every class, it effectively controls the data distribution for most classes, thereby mitigating the limitations imposed by rare categories.

7) Evaluation of Cross-Dataset Generalization: To evaluate the effectiveness of the proposed synthetic data in cross-dataset generalization, we trained the YOLOv8 on both the DIOR-R dataset and its augmented DIOR-R dataset. The trained model was then tested in a cross-domain setting on the NWPU VHR-10 dataset, without using any data from the NWPU VHR-10 dataset during the training process. As shown in Table XV, synthetic data consistently enhances cross-domain performance. Categories containing small objects, such as *vehicle*, benefit the most, while categories that already achieve high accuracy, such as *airplane* and *basketball court*, exhibit stable or slightly improved results. The overall mAP50 increases from 69.3 without synthetic data to 70.6 with 20K synthetic samples, demonstrating that the synthesized data possesses strong cross-dataset generalization capability.

C. Comparison With State of the Arts

To assess image generation quality, we compare LSDGen with several state-of-the-art L2I generation methods, including LostGAN [33], ReCo [34], LayoutDiffusion [10], GLIGEN [35], and AeroGen [2]. The FID scores reported in Table V demonstrate that LSDGen achieves the best visual fidelity among all competing approaches, indicating the effectiveness of LSDGen in producing high-quality remote sensing imagery.

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TABLE XI

DETECTION PERFORMANCE AND DISTRIBUTION COMPARISON WITH VARYING SYNTHETIC DATA SCALES ON DIOR-R

Class	Real Distribution		Synthetic Distribution (10K)		Synthetic Distribution (20K)		Synthetic Distribution (50K)	
	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP
all	–	66.3 / 49.3	–	67.3 / 51.3	–	68.1 / 52.0	–	68.5 / 52.5
airplane	1157 (3.61%)	84.9 / 67.4	624 (2.47%)	83.3 / 68.1	1310 (2.63%)	83.8 / 68.8	3305 (2.67%)	84.0 / 69.2
airport	316 (0.99%)	44.8 / 18.1	434 (1.72%)	46.4 / 17.8	846 (1.70%)	48.8 / 18.9	2064 (1.66%)	49.2 / 19.3
baseballfield	1216 (3.79%)	78.8 / 71.4	896 (3.55%)	72.7 / 68.3	1784 (3.59%)	74.0 / 69.3	4413 (3.56%)	74.4 / 69.8
basketballcourt	551 (1.72%)	89.8 / 80.5	614 (2.44%)	91.2 / 84.4	1165 (2.34%)	91.3 / 84.7	2877 (2.32%)	91.4 / 84.9
bridge	491 (1.53%)	38.9 / 22.9	164 (0.65%)	39.7 / 24.1	336 (0.68%)	40.7 / 24.8	1172 (0.95%)	41.5 / 25.4
chimney	332 (1.04%)	76.9 / 66.2	372 (1.48%)	79.0 / 70.3	729 (1.47%)	79.5 / 70.8	1844 (1.49%)	79.7 / 71.1
dam	251 (0.78%)	29.0 / 12.2	382 (1.52%)	36.1 / 15.2	813 (1.63%)	36.3 / 15.5	1891 (1.53%)	37.0 / 16.0
Expressway-Service-area	431 (1.34%)	47.7 / 33.8	133 (0.53%)	49.8 / 36.5	276 (0.55%)	50.4 / 36.9	682 (0.55%)	50.8 / 37.4
Expressway-toll-station	292 (0.91%)	57.0 / 48.6	267 (1.06%)	56.6 / 49.1	549 (1.10%)	57.4 / 49.8	1394 (1.12%)	58.0 / 50.4
golffield	232 (0.72%)	73.7 / 48.6	359 (1.42%)	77.6 / 56.2	703 (1.41%)	77.3 / 56.6	1813 (1.46%)	77.7 / 57.3
groundtrackfield	530 (1.65%)	64.1 / 52.1	748 (2.97%)	64.6 / 54.5	1465 (2.95%)	65.6 / 55.4	3631 (2.93%)	65.8 / 55.8
harbor	1013 (3.16%)	54.8 / 31.8	819 (3.25%)	55.2 / 35.1	1536 (3.09%)	55.8 / 35.6	3710 (2.99%)	56.3 / 35.9
overpass	565 (1.76%)	56.9 / 38.3	465 (1.84%)	60.7 / 41.4	971 (1.95%)	61.1 / 42.0	2359 (1.90%)	61.5 / 42.0
ship	12982 (40.51%)	92.3 / 75.4	1869 (7.41%)	91.9 / 75.2	3706 (7.45%)	92.1 / 75.6	9265 (7.47%)	92.2 / 76.0
stadium	272 (0.85%)	71.3 / 56.4	942 (3.74%)	73.2 / 59.5	1872 (3.76%)	74.7 / 60.7	4605 (3.71%)	75.5 / 61.4
storagetank	1337 (4.17%)	82.6 / 65.3	1470 (5.83%)	78.7 / 64.5	3087 (6.21%)	79.5 / 65.3	7563 (6.10%)	80.1 / 65.8
tenniscourt	2226 (6.95%)	89.2 / 83.7	2762 (10.95%)	90.9 / 86.3	5636 (11.33%)	91.1 / 86.8	13526 (10.91%)	91.2 / 87.0
trainstation	246 (0.77%)	57.4 / 29.7	158 (0.63%)	66.4 / 37.6	336 (0.68%)	67.9 / 38.7	853 (0.69%)	68.9 / 40.1
vehicle	6646 (20.74%)	55.0 / 38.2	10282 (40.78%)	55.1 / 38.4	19583 (39.38%)	56.0 / 39.2	49542 (39.95%)	56.5 / 39.6
windmill	961 (3.00%)	80.4 / 45.3	1453 (5.76%)	77.6 / 43.6	3031 (6.09%)	78.7 / 44.6	7486 (6.04%)	79.0 / 44.9

TABLE XII

DETECTION PERFORMANCE AND DISTRIBUTION COMPARISON WITH VARYING SYNTHETIC DATA SCALES ON DIOR

Class	Real Distribution		Synthetic Distribution (10K)		Synthetic Distribution (20K)		Synthetic Distribution (50K)	
	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP
all	—	74.4 / 54.1	—	75.1 / 55.0	—	76.7 / 56.2	—	77.8 / 57.0
airplane	1157 (3.61%)	91.3 / 66.4	624 (2.47%)	74.9 / 55.2	1310 (2.63%)	75.1 / 56.7	3305 (2.67%)	76.4 / 57.3
airport	316 (0.99%)	78.4 / 55.8	434 (1.72%)	85.4 / 63.3	846 (1.70%)	87.0 / 64.5	2064 (1.66%)	88.1 / 65.9
baseballfield	1216 (3.79%)	81.7 / 67.0	896 (3.55%)	80.1 / 68.7	1784 (3.59%)	81.8 / 70.2	4413 (3.56%)	82.8 / 71.4
basketballcourt	551 (1.72%)	91.2 / 80.9	614 (2.44%)	92.3 / 82.0	1165 (2.34%)	94.4 / 83.8	2877 (2.32%)	96.4 / 84.7
bridge	491 (1.53%)	46.4 / 26.8	164 (0.65%)	50.1 / 29.5	336 (0.68%)	51.6 / 30.3	1172 (0.95%)	51.9 / 30.5
chimney	332 (1.04%)	80.2 / 68.5	372 (1.48%)	83.3 / 71.0	729 (1.47%)	84.7 / 72.3	1844 (1.49%)	86.5 / 73.2
dam	251 (0.78%)	60.3 / 33.0	382 (1.52%)	66.1 / 37.5	813 (1.63%)	68.0 / 38.2	1891 (1.53%)	68.5 / 38.8
Expressway-Service-area	431 (1.34%)	63.7 / 49.6	133 (0.53%)	67.0 / 52.7	276 (0.55%)	68.8 / 54.3	682 (0.55%)	69.7 / 55.1
Expressway-toll-station	292 (0.91%)	62.4 / 51.0	267 (1.06%)	69.0 / 55.0	549 (1.10%)	70.9 / 56.1	1394 (1.12%)	71.7 / 56.7
golffield	232 (0.72%)	78.6 / 59.7	359 (1.42%)	84.4 / 66.9	703 (1.41%)	86.3 / 68.3	1813 (1.46%)	87.4 / 69.4
groundtrackfield	530 (1.65%)	80.0 / 63.6	748 (2.97%)	76.7 / 59.4	1465 (2.95%)	77.8 / 60.7	3631 (2.93%)	79.3 / 61.6
harbor	1013 (3.16%)	64.4 / 45.5	819 (3.25%)	66.5 / 47.8	1536 (3.09%)	67.9 / 49.1	3710 (2.99%)	68.9 / 49.8
overpass	565 (1.76%)	62.1 / 43.4	465 (1.84%)	63.6 / 44.5	971 (1.95%)	64.8 / 45.5	2359 (1.90%)	65.7 / 46.2
ship	12982 (40.51%)	94.6 / 63.9	1869 (7.41%)	92.9 / 63.1	3706 (7.45%)	94.2 / 64.6	9265 (7.47%)	95.7 / 65.4
stadium	272 (0.85%)	76.2 / 57.7	942 (3.74%)	69.4 / 49.0	1872 (3.76%)	71.1 / 49.6	4605 (3.71%)	71.9 / 50.8
storagetank	1337 (4.17%)	86.7 / 59.3	1470 (5.83%)	81.5 / 55.8	3087 (6.21%)	82.9 / 56.8	7563 (6.10%)	84.4 / 57.7
tenniscourt	2226 (6.95%)	91.6 / 81.4	2762 (10.95%)	90.6 / 80.4	5636 (11.33%)	92.5 / 82.1	13526 (10.91%)	93.7 / 83.2
trainstation	246 (0.77%)	54.9 / 28.8	158 (0.63%)	66.8 / 37.5	336 (0.68%)	68.2 / 38.4	853 (0.69%)	69.5 / 38.8
vehicle	6646 (20.74%)	61.9 / 36.3	10282 (40.78%)	60.9 / 36.5	19583 (39.38%)	62.7 / 37.5	49542 (39.95%)	63.3 / 38.0
windmill	961 (3.00%)	82.3 / 43.0	1453 (5.76%)	81.0 / 44.2	3031 (6.09%)	82.7 / 45.2	7486 (6.04%)	84.3 / 45.6

TABLE XIII

COMPARISON OF DETECTION PERFORMANCE USING ORIGINAL AND REBALANCED SYNTHETIC DATA DISTRIBUTIONS ON THE DIOR-R DATASET

Class	Original Synthetic Distribution (10K)		Rebalanced Synthetic Distribution (10K)		Original Synthetic Distribution (20K)		Rebalanced Synthetic Distribution (20K)	
	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP
all	—	67.3 / 51.3	—	67.5 / 52.1	—	68.1 / 52.0	—	68.5 / 53.1
airplane	624 (2.47%)	83.3 / 68.1	808 (4.37%)	84.5 / 71.8	1310 (2.63%)	83.8 / 68.8	1454 (4.16%)	85.3 / 73.0
airport	434 (1.72%)	46.4 / 17.8	503 (2.72%)	41.0 / 15.8	846 (1.70%)	48.8 / 18.9	1021 (2.92%)	43.3 / 17.1
baseballfield	896 (3.55%)	72.7 / 68.3	617 (3.34%)	76.1 / 72.5	1784 (3.59%)	74.0 / 69.3	1283 (3.67%)	76.5 / 73.0
basketballcourt	614 (2.44%)	91.2 / 84.4	659 (3.57%)	91.7 / 84.6	1165 (2.34%)	91.3 / 84.7	1302 (3.73%)	92.0 / 85.6
bridge	164 (0.65%)	39.7 / 24.1	638 (3.45%)	42.3 / 26.4	336 (0.68%)	40.7 / 24.8	1244 (3.56%)	43.8 / 27.2
chimney	372 (1.48%)	79.0 / 70.3	823 (4.46%)	84.5 / 76.8	729 (1.47%)	79.5 / 70.8	1481 (4.24%)	85.0 / 77.5
dam	382 (1.52%)	36.1 / 15.2	500 (2.71%)	35.7 / 15.6	813 (1.63%)	36.3 / 15.5	1000 (2.86%)	37.8 / 16.7
Expressway-Service-area	133 (0.53%)	49.8 / 36.5	522 (2.83%)	51.0 / 36.7	276 (0.55%)	50.4 / 36.9	1059 (3.03%)	52.5 / 38.0
Expressway-toll-station	267 (1.06%)	56.6 / 49.1	517 (2.80%)	58.7 / 51.9	549 (1.10%)	57.4 / 49.8	1085 (3.10%)	59.9 / 53.0
golffield	359 (1.42%)	77.6 / 56.2	504 (2.73%)	76.4 / 56.6	703 (1.41%)	77.3 / 56.6	1000 (2.86%)	78.0 / 58.3
groundtrackfield	748 (2.97%)	64.6 / 54.5	617 (3.34%)	64.3 / 55.0	1465 (2.95%)	65.6 / 55.4	1271 (3.64%)	65.2 / 56.1
harbor	819 (3.25%)	55.2 / 35.1	539 (2.92%)	52.6 / 31.6	1536 (3.09%)	55.8 / 35.6	1067 (3.05%)	53.6 / 32.7
overpass	465 (1.84%)	60.7 / 41.4	882 (4.78%)	62.6 / 43.1	971 (1.95%)	61.1 / 42.0	1746 (5.00%)	62.9 / 43.6
ship	1869 (7.41%)	91.9 / 75.2	827 (4.48%)	91.6 / 74.9	3706 (7.45%)	92.1 / 75.6	1620 (4.64%)	91.9 / 75.5
stadium	942 (3.74%)	73.2 / 59.5	558 (3.02%)	64.8 / 53.4	1872 (3.76%)	74.7 / 60.7	1131 (3.24%)	66.6 / 54.3
storagetank	1470 (5.83%)	78.7 / 64.5	532 (2.88%)	78.7 / 65.0	3087 (6.21%)	79.5 / 65.3	1170 (3.35%)	79.2 / 65.8
tenniscourt	2762 (10.95%)	90.9 / 86.3	1198 (6.49%)	91.4 / 87.5	5636 (11.33%)	91.1 / 86.8	2503 (7.16%)	91.7 / 88.0
trainstation	158 (0.63%)	66.4 / 37.6	500 (2.71%)	68.0 / 37.1	336 (0.68%)	67.9 / 38.7	1000 (2.86%)	68.9 / 38.8
vehicle	10282 (40.78%)	55.1 / 38.4	6089 (32.97%)	52.8 / 38.4	19583 (39.38%)	56.0 / 39.2	10351 (29.62%)	54.0 / 39.3
windmill	1453 (5.76%)	77.6 / 43.6	638 (3.45%)	81.7 / 48.2	3031 (6.09%)	78.7 / 44.6	1163 (3.33%)	82.3 / 49.0

We further compare LSDGen with conventional data augmentation techniques on the DIOR-R dataset, including Flip [37] and copy–paste [37]. As shown in Table VI, LSDGen significantly outperforms these methods in improving the

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Fig. 7. Inference results of YOLOv8 on the DIOR dataset.

performance of object detection. Unlike traditional augmentations that rely solely on geometric or compositional transformations, LSDGen synthesizes semantically coherent layouts, enhancing both visual quality and scene diversity.

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Fig. 8. Inference results of YOLOv8 on the DIOR-R dataset.

Comprehensive evaluations conducted on the DIOR, DIOR-R, and HRSC datasets (see Table VII) further substantiate that LSDGen either compares with or outperforms AeroGen in improving object detection performance. This

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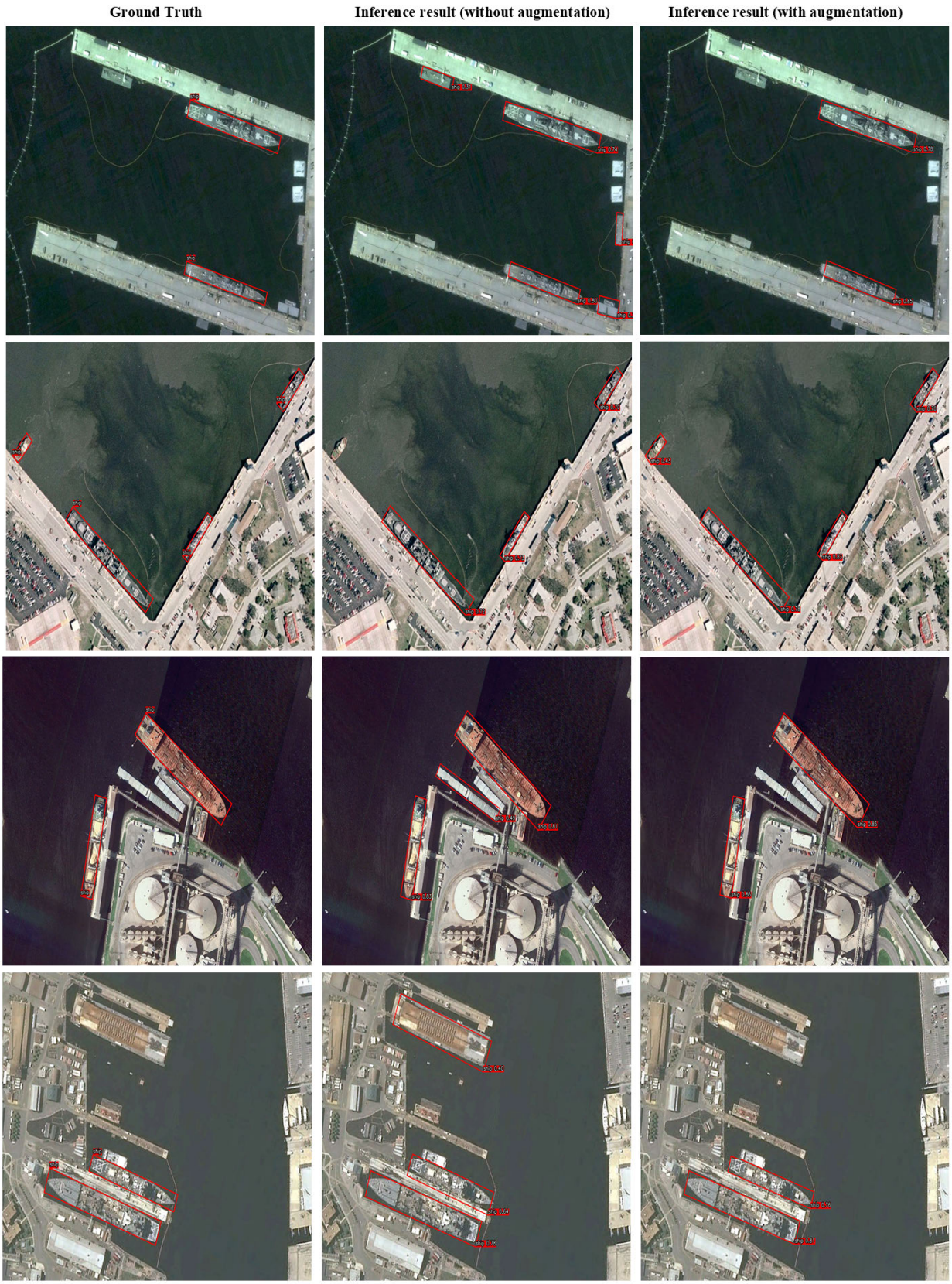


Fig. 9. Inference results of YOLOv8 on the HRSC dataset.

performance gain primarily results from the integration of the RLGen module, which leverages category co-occurrence information to construct a category association matrix for generating diverse and semantically coherent layouts, and the

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TABLE XIV

COMPARISON OF DETECTION PERFORMANCE USING ORIGINAL AND REBALANCED SYNTHETIC DATA DISTRIBUTIONS ON THE DIOR DATASET

Class	Original Synthetic Distribution (10K)		Rebalanced Synthetic Distribution (10K)		Original Synthetic Distribution (20K)		Rebalanced Synthetic Distribution (20K)	
	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP	Count (%)	mAP50 / mAP
all	–	75.1 / 55.0	–	74.9 / 55.8	–	76.7 / 56.2	–	76.4 / 56.3
airplane	624 (2.47%)	74.9 / 55.2	808 (4.37%)	83.7 / 62.6	1310 (2.63%)	75.1 / 56.7	1454 (4.16%)	84.1 / 62.5
airport	434 (1.72%)	85.4 / 63.3	503 (2.72%)	82.1 / 61.5	846 (1.70%)	87.0 / 64.5	1021 (2.92%)	82.4 / 61.8
baseballfield	896 (3.55%)	80.1 / 68.7	617 (3.34%)	79.1 / 69.7	1784 (3.59%)	81.8 / 70.2	1283 (3.67%)	80.6 / 70.6
basketballcourt	614 (2.44%)	92.3 / 82.0	659 (3.57%)	92.0 / 82.6	1165 (2.34%)	94.4 / 83.8	1302 (3.73%)	94.2 / 83.2
bridge	164 (0.65%)	50.1 / 29.5	638 (3.45%)	50.3 / 29.8	336 (0.68%)	51.6 / 30.3	1244 (3.56%)	51.6 / 30.2
chimney	372 (1.48%)	83.3 / 71.0	823 (4.46%)	82.5 / 72.3	729 (1.47%)	84.7 / 72.3	1481 (4.24%)	85.3 / 73.1
dam	382 (1.52%)	66.1 / 37.5	500 (2.71%)	69.7 / 39.3	813 (1.63%)	68.0 / 38.2	1000 (2.86%)	69.0 / 38.6
Expressway-Service-area	133 (0.53%)	67.0 / 52.7	522 (2.83%)	66.0 / 53.1	276 (0.55%)	68.8 / 54.3	1059 (3.03%)	69.8 / 55.4
Expressway-toll-station	267 (1.06%)	69.0 / 55.0	517 (2.80%)	66.6 / 56.7	549 (1.10%)	70.9 / 56.1	1085 (3.10%)	70.2 / 59.1
golffield	359 (1.42%)	84.4 / 66.9	504 (2.73%)	82.7 / 66.3	703 (1.41%)	86.3 / 68.3	1000 (2.86%)	83.6 / 66.5
groundtrackfield	748 (2.97%)	76.7 / 59.4	617 (3.34%)	77.4 / 61.4	1465 (2.95%)	77.8 / 60.7	1271 (3.64%)	77.4 / 60.4
harbor	819 (3.25%)	66.5 / 47.8	539 (2.92%)	65.6 / 47.2	1536 (3.09%)	67.9 / 49.1	1067 (3.05%)	67.1 / 47.6
overpass	465 (1.84%)	63.6 / 44.5	882 (4.78%)	63.8 / 45.0	971 (1.95%)	64.8 / 45.5	1746 (5.00%)	65.8 / 45.8
ship	1869 (7.41%)	92.9 / 63.1	827 (4.48%)	92.9 / 64.1	3706 (7.45%)	94.2 / 64.6	1620 (4.64%)	94.7 / 64.3
stadium	942 (3.74%)	69.4 / 49.0	558 (3.02%)	64.8 / 47.6	1872 (3.76%)	71.1 / 49.6	1131 (3.24%)	63.7 / 45.6
storagetank	1470 (5.83%)	81.5 / 55.8	532 (2.88%)	80.6 / 55.2	3087 (6.21%)	82.9 / 56.8	1170 (3.35%)	82.0 / 55.5
tenniscourt	2762 (10.95%)	90.6 / 80.4	1198 (6.49%)	91.1 / 82.2	5636 (11.33%)	92.5 / 82.1	2503 (7.16%)	92.3 / 82.7
trainstation	158 (0.63%)	66.8 / 37.5	500 (2.71%)	65.0 / 36.7	336 (0.68%)	68.2 / 38.4	1000 (2.86%)	68.8 / 38.6
vehicle	10282 (40.78%)	60.9 / 36.5	6089 (32.97%)	60.3 / 37.1	19583 (39.38%)	62.7 / 37.5	10351 (29.62%)	61.6 / 37.5
windmill	1453 (5.76%)	81.0 / 44.2	638 (3.45%)	81.0 / 45.3	3031 (6.09%)	82.7 / 45.2	1163 (3.33%)	84.0 / 46.8

TABLE XV

PERFORMANCE COMPARISON ON THE NWPU VHR-10 DATASET

Class	DIOR-R data	+10K	+20K	+50K
all	69.3 / 42.9	69.8 / 43.8	70.6 / 43.8	70.4 / 43.7
airplane	95.5 / 74.1	96.6 / 74.6	96.6 / 74.4	96.9 / 75.1
ship	1.88 / 0.39	1.90 / 0.50	2.18 / 0.51	2.10 / 0.51
storage tank	97.9 / 51.6	98.6 / 56.6	99.1 / 54.3	98.8 / 55.8
baseball diamond	98.2 / 78.5	97.8 / 80.0	97.9 / 80.5	97.6 / 80.0
tennis court	93.3 / 51.0	92.1 / 49.4	91.8 / 48.0	92.5 / 49.6
basketball court	92.9 / 60.1	92.3 / 62.1	91.8 / 60.8	94.5 / 63.1
ground track field	97.0 / 69.1	94.9 / 63.2	95.9 / 65.4	95.1 / 63.1
harbor	75.9 / 29.3	70.4 / 29.4	74.4 / 31.4	72.9 / 27.7
bridge	5.37 / 1.08	6.97 / 1.28	5.65 / 1.98	6.37 / 1.07
vehicle	35.3 / 13.8	46.7 / 20.7	50.9 / 21.7	47.0 / 21.3

ISDF module, which filters low-quality synthetic data to yield higher-quality training samples for the detector. Fig. 6 presents a qualitative comparison of layouts generated by our method and AeroGen, together with the remote sensing images synthesized from these layouts. This visualization demonstrates the superior diversity and structural plausibility of our approach, leading to high-quality image synthesis.

D. Qualitative Results

Figs. 7–9, respectively, illustrate the visualization of YOLOv8 on the test sets of DIOR, DIOR-R, and HRSC. The results demonstrate that the synthetic data generated by our method can enhance the detector's capability in identifying small objects and rare categories, thereby validating the effectiveness of our approach. This improvement is attributed to our method's ability to generate diverse, high-quality data while effectively filtering out low-quality data, which, in turn, boosts detection performance.

E. Overall Complexity

We further evaluate the computational overhead of the three components of our method: RLGen, LCDM, and ISDF. As presented in Table XVI, RLGen demonstrates high efficiency, requiring only approximately 0.006 seconds per layout. The LCDM stage, however, is the dominant cost factor, taking around 8 seconds to synthesize a 512×512 image. In contrast, the ISDF stage is significantly faster, processing each image

TABLE XVI

COMPUTATIONAL COST OF THE THREE STAGES IN THE PROPOSED PIPELINE, MEASURED ON A SINGLE NVIDIA RTX 3090 GPU WITH 24 GB MEMORY

Component	Average Cost per Sample	Cost for 10,000 Samples
RLGen	0.006 seconds	1 minutes
LCDM	8 seconds	23 hours
ISDF	0.03-0.04 seconds	6 minutes

in approximately 0.03–0.04 s. Overall, the complete pipeline takes approximately 24 h to generate 10 000 synthetic samples.

V. CONCLUSION

To enhance the performance of remote sensing object detection, we propose LSDGen, a novel framework for generating high-quality synthetic training data. The LSDGen is composed of three key components. First, we introduce RLGen, which leverages a matrix of intercategory correlation statistics mined from real-world datasets to guide the layout generation process. This approach facilitates the generation of semantically coherent layouts and effectively mitigates the semantic conflicts commonly associated with purely random generation. Second, the generated layouts from RLGen are used to condition the LCDM to synthesize corresponding remote sensing images. Finally, to further enhance the performance of the synthetic images in downstream detection tasks, we apply ISDF to remove noisy or low-confidence samples, using semantic consistency measured via CLIP-based visual–textual similarity. Extensive experiments demonstrate that our framework significantly boosts detection performance across multiple benchmarks, highlighting the effectiveness and scalability of layout-guided synthetic data generation for remote sensing image object detection.

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